EGLE

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY
WATER RESOURCES DIVISION

BIOSOLIDS LAND APPLICATION SITE IDENTIFICATION FORM

Authorized under 1994 PA 451, as amended. Completion of form is required. Applicant may be subject to civil and/or criminal penalties for providing false information.

Complete the following information and provide the attachments listed on the second page for each new or updated biosolids land application site. Please amend any previous Site Identification Forms when it becomes known that any of the information requested below has changed (i.e., new landowner).

For additional information, Biosolids Program Staff are listed at: https://www.michigan.gov/documents/deq/wrd-biosolids-staff_402800_7.pdf.

Facility Name Cadillac WWTP	NPDES Permit or COC Number MI0020257	Request: New Site / Site Update / Site transfer New Site

Site ID Name (ex T/R/S – Owner (05N08E03-LB01) T22NR11W-RBL02	Site location description (street name, nearest crossroad) W. 36 Rd E. of S. 23 RD
County Wexford	Township Boon

(ex: NW, NE, SW, SE) SE 1/4	(ex: NW, NE, SW, SE) SW 1/4	Township (ex: 05N) T22N	Range (ex: 08E) R11W	Section 25

Latitude (ex: N 42.09651) 44.1606 N	Longitude (ex: -81.09651) 85.3525 W	Method Used (i.e., GPS, topo) Google Earth

Name of Owner Rick & Barbara Liptak	Phone		
Street or P.O. Box 621 W. 36 Rd	City or Town Boon	State MI	Zip 49618

Is there a written agreement from the owner to land apply?
- ☒ Yes
- ☐ No (written agreement is required)

Previous Landowner's Name (if site update)	Property Purchase Date		
Name of Lessee/Farmer (if different) Fenner Farms	Phone		
Street or P.O. Box 2601 S. 23 Rd	City or Town Boon	State MI	Zip 49618

Site Type
- ☒ Agricultural
- ☐ Forest
- ☐ *Reclamation Site (separate approval is required)
- ☐ Lawn/Home Garden (requires EQ Biosolids)

Soils Information

Date of last soils analysis 8/17/2022

*Phosphorous 79 units PPM

Potassium 89 units PPM

pH 5.8 Soil type(s) Sandy

*300 lb/ac (or 150 ppm) Bray P1 maximum limit

Percent slope

Highest percent slope of the site?
- ☐ 0-6%
- ☒ 6-12%
- ☐ *Higher (Requires an approved site reclamation plan)

Expected Crops/Vegetation to be grown

Crop/Vegetation: corn	Nitrogen Requirement: 135 lbs. N/acre
Crop/Vegetation: ____	Nitrogen Requirement: ____ lbs. N/acre
Crop/Vegetation: ____	Nitrogen Requirement: ____ lbs. N/acre

Total acreage of site 80	**Acreage used for crops** 25	**Acreage used for land application**

Does this site have a high potential for public exposure?
- ☐ Yes
- ☑ No

If agricultural, is the site tiled?
- ☐ Yes
- ☐ No
- ☑ NA

CITY OF CADILLAC MICHIGAN

PARCEL A ABL 02

Cadillac Utilities Department

CITY OF CADILLAC WASTEWATER TREATMENT PLANT

BIOSOLIDS RECYCLING PROGRAM PARTICIPATION

I hereby agree to participate in the City of Cadillac WWTP biosolids recycling program. I understand that biosolids will be applied to my land in accordance with federal EPA regulations and guidelines of the Michigan Department of Environmental Quality.

I understand that biosolids have been analyzed to be sure they can be applied to agricultural land in accordance with federal regulations and MDEQ guidelines. I understand that biosolids contain some heavy metals and that they must be below the EPA allowable limits for land application. I understand that the nutrients and/or the metal concentration may limit the frequency of biosolids application to my land.

My land is used for the production of agricultural crops. Information concerning crop and site restrictions has been explained to me. I agree to allow representatives of the MDEQ, Cadillac Waste Water Treatment Plant and the City's Injection contractor to have access to my land to obtain soil samples and for application of biosolids.

If I decide to cancel this agreement, I will give 60 days advance notice to the City of Cadillac WWTP.

Landowner: RICK & BARBARA LIPTAK Signed: Rick [signature] / Barbara [signature]

Address: 621 W 36 RD Date: FEB 28, 2022

Township: BOON Section: 25 Acres: 25

City WWTP Field Number: T22N R11W RBL02

If Rented, Farmer: LEASE FENNER FARMS

Address: 2601 S 23 RD BOON 49618

phone owner 231-429-2425
Farmer 231-775-3276

Provide copies of the following as attachments:

- USDA Soil Survey Map of the application site with soils information, property boundaries, surface waters, and location of field tiles when present.
- Most recent biosolids analytical data.
- Soil fertility analysis (no older than 2 years old at the time of land application).
- Plat map (or other map showing property dimensions) with the location of the land application site and street names clearly highlighted.
- Signed landowner / farmer agreements.
- Township / County Health Department notification.

For information or assistance on this publication, please contact the Biosolids Program, through EGLE's Environmental Assistance Center at 800-662-9278. This publication is available in alternative formats upon request.

BOON T22N • R11W

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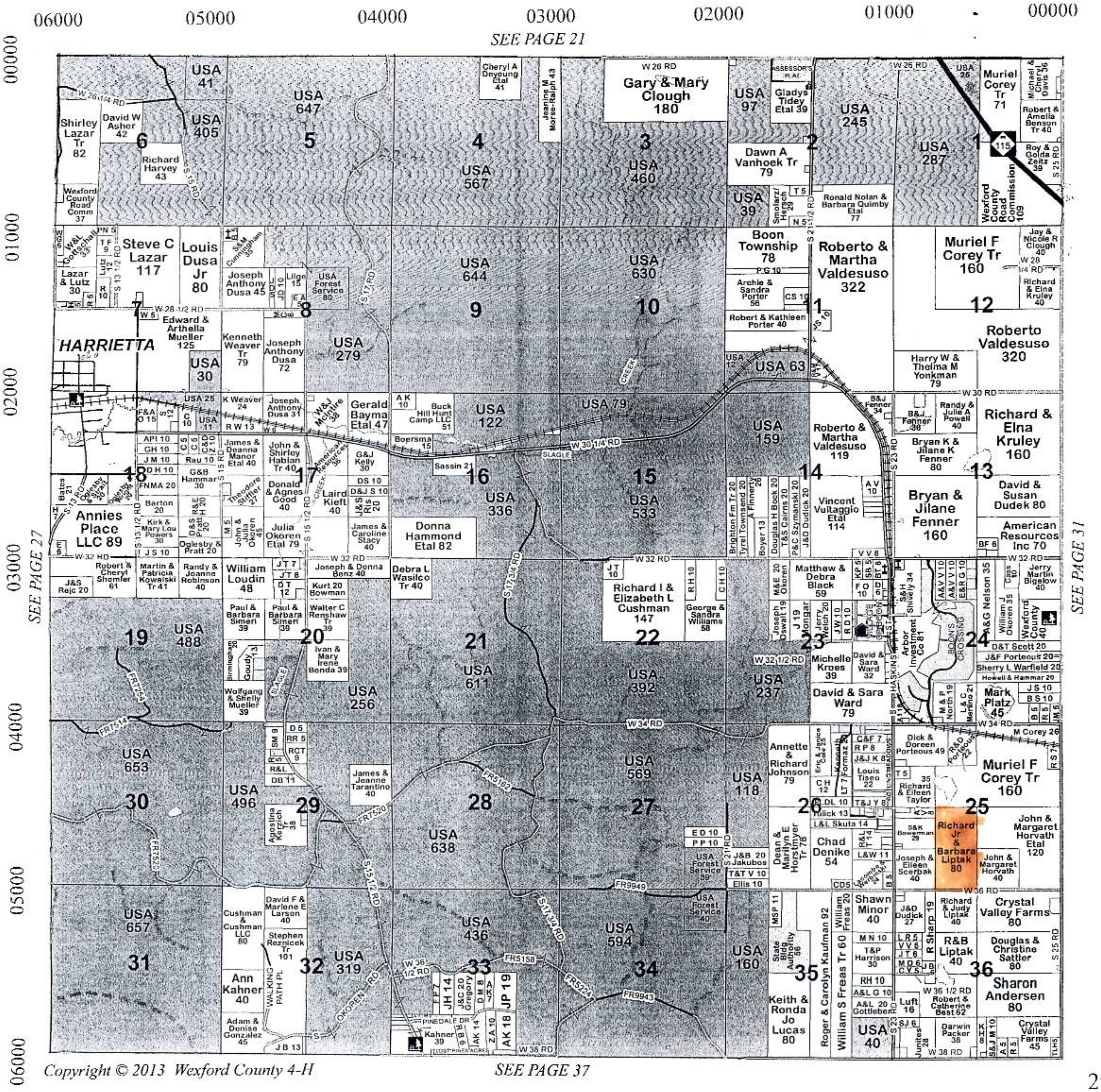

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MICHIGAN STATE UNIVERSITY

MICHIGAN STATE UNIVERSITY
SOIL AND PLANT NUTRIENT LABORATORY
EAST LANSING, MICHIGAN 48824-1325
(517) 355-0218

SOIL TEST REPORT FOR:	CONSULTANT
CITY OF CADILLAC 200 N. LAKE ST. CADILLAC MI 49601	WEXFORD COUNTY MSUE 9600 E. 13TH ST. CADILLAC MI 49601 231-779-9480

DATE	LAB #	COUNTY	Previous Crop	ACRES	FIELD ID	SOIL	TEXTURE
8-17-2022	269325	Wexford		15	T22nr11wrbl02	Mineral	

SOIL NUTRIENT LEVELS			Below Optimum	Optimum	Above Optimum
[1]Soil pH 5.8	Lime Index 69.0				
[2]Phosphorus (P)	79	ppm	■■■■■■■■■■	■■■■■■■■■■	■■■■■■■■■■■■■■■■■■■■
[3]Potassium (K)	89	ppm	■■■■■■■■■■	■	
[3]Magnesium (Mg)	104	ppm	■■■■■■■■■■	■■■■■■■■■■	■■■■■

ADDITIONAL RESULTS:					Optional Tests:						
Calcium (Ca) (ppm)	CEC (meq/100 g)	% of Exchangeable Bases			Micronutrients (ppm)					Organic Matter %	Nitrate-N ppm
		K	Mg	Ca	B	Cu	Mn	Zn	Fe		
825	6.4	4.4	16.6	79.0							

RECOMMENDATIONS:

Limestone: 2 ton/A

Target pH = 6.5

Tillage Depth: 9 inches

% Stand:

Plant Nutrients:

Year Crop	Expected Yield	Nitrogen (lb N/A)	Phosphate (lb P_2O_5/A)	Potassium (lb K_2O/A)	Boron (lb B/A)	Manganese (lb Mn/A)	Zinc (lb Zn/A)	Copper (lb Cu/A)
					Micronutrient: (Optional)			
1 Corn silage	18 ton	135	0	145	0.0			

Use a starter fertilizer high in nitrogen (30-40 lb N/A)) when planting into heavy residue.
This is a weakly buffered soil, so the lime recommendation is based on soil pH and target pH for the first year crop.

For questions about interpreting your soil results go to the following website: http://msue.anr.msu.edu/experts and contact the MSUE field crops agent in your area.

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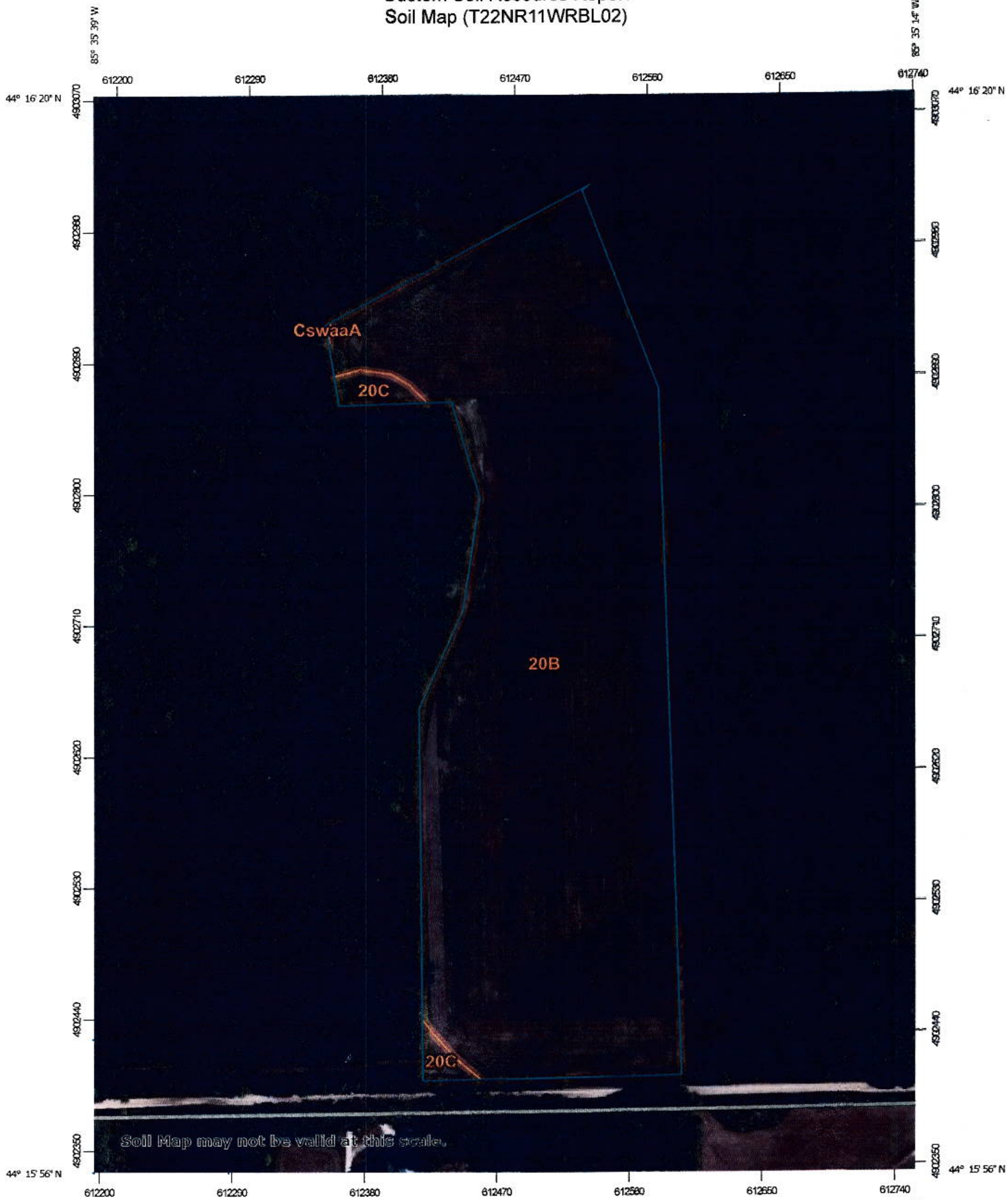
Test Methods: 1- 1:1 soil:water pH, 2- Bray P1 Extractant, 3- 1N Ammonium Acetate Extractant

Custom Soil Resource Report
Soil Map (T22NR11WRBL02)



Map Scale: 1:3,590 if printed on A portrait (8.5" x 11") sheet.

Map projection: Web Mercator Corner coordinates: WGS84 Edge tics: UTM Zone 16N WGS84

MAP LEGEND

Area of Interest (AOI)
- Area of Interest (AOI)

Soils
- Soil Map Unit Polygons
- Soil Map Unit Lines
- Soil Map Unit Points

Special Point Features
- Blowout
- Borrow Pit
- Clay Spot
- Closed Depression
- Gravel Pit
- Gravelly Spot
- Landfill
- Lava Flow
- Marsh or swamp
- Mine or Quarry
- Miscellaneous Water
- Perennial Water
- Rock Outcrop
- Saline Spot
- Sandy Spot
- Severely Eroded Spot
- Sinkhole
- Slide or Slip
- Sodic Spot
- Spoil Area
- Stony Spot
- Very Stony Spot
- Wet Spot
- Other
- Special Line Features

Water Features
- Streams and Canals

Transportation
- Rails
- Interstate Highways
- US Routes
- Major Roads
- Local Roads

Background
- Aerial Photography

MAP INFORMATION

The soil surveys that comprise your AOI were mapped at 1:15,800.

Warning: Soil Map may not be valid at this scale.

Enlargement of maps beyond the scale of mapping can cause misunderstanding of the detail of mapping and accuracy of soil line placement. The maps do not show the small areas of contrasting soils that could have been shown at a more detailed scale.

Please rely on the bar scale on each map sheet for map measurements.

Source of Map: Natural Resources Conservation Service
Web Soil Survey URL:
Coordinate System: Web Mercator (EPSG:3857)

Maps from the Web Soil Survey are based on the Web Mercator projection, which preserves direction and shape but distorts distance and area. A projection that preserves area, such as the Albers equal-area conic projection, should be used if more accurate calculations of distance or area are required.

This product is generated from the USDA-NRCS certified data as of the version date(s) listed below.

Soil Survey Area: Lake and Wexford Counties, Michigan
Survey Area Data: Version 15, Sep 1, 2021

Soil map units are labeled (as space allows) for map scales 1:50,000 or larger.

Date(s) aerial images were photographed: Jul 2, 2020—Nov 12, 2020

The orthophoto or other base map on which the soil lines were compiled and digitized probably differs from the background imagery displayed on these maps. As a result, some minor shifting of map unit boundaries may be evident.

Map Unit Legend (T22NR11WRBL02)

Map Unit Symbol	Map Unit Name	Acres in AOI	Percent of AOI
20B	Montcalm-Graycalm complex, 0 to 6 percent slopes	22.9	98.0%
20C	Montcalm-Graycalm complex, 6 to 12 percent slopes	0.5	2.0%
CswaaA	Croswell sand, 0 to 6 percent slopes	0.0	0.0%
Totals for Area of Interest		**23.4**	**100.0%**

Map Unit Descriptions (T22NR11WRBL02)

The map units delineated on the detailed soil maps in a soil survey represent the soils or miscellaneous areas in the survey area. The map unit descriptions, along with the maps, can be used to determine the composition and properties of a unit.

A map unit delineation on a soil map represents an area dominated by one or more major kinds of soil or miscellaneous areas. A map unit is identified and named according to the taxonomic classification of the dominant soils. Within a taxonomic class there are precisely defined limits for the properties of the soils. On the landscape, however, the soils are natural phenomena, and they have the characteristic variability of all natural phenomena. Thus, the range of some observed properties may extend beyond the limits defined for a taxonomic class. Areas of soils of a single taxonomic class rarely, if ever, can be mapped without including areas of other taxonomic classes. Consequently, every map unit is made up of the soils or miscellaneous areas for which it is named and some minor components that belong to taxonomic classes other than those of the major soils.

Most minor soils have properties similar to those of the dominant soil or soils in the map unit, and thus they do not affect use and management. These are called noncontrasting, or similar, components. They may or may not be mentioned in a particular map unit description. Other minor components, however, have properties and behavioral characteristics divergent enough to affect use or to require different management. These are called contrasting, or dissimilar, components. They generally are in small areas and could not be mapped separately because of the scale used. Some small areas of strongly contrasting soils or miscellaneous areas are identified by a special symbol on the maps. If included in the database for a given area, the contrasting minor components are identified in the map unit descriptions along with some characteristics of each. A few areas of minor components may not have been observed, and consequently they are not mentioned in the descriptions, especially where the pattern was so complex that it was impractical to make enough observations to identify all the soils and miscellaneous areas on the landscape.

The presence of minor components in a map unit in no way diminishes the usefulness or accuracy of the data. The objective of mapping is not to delineate pure taxonomic classes but rather to separate the landscape into landforms or